Configuration: Data Scenarios

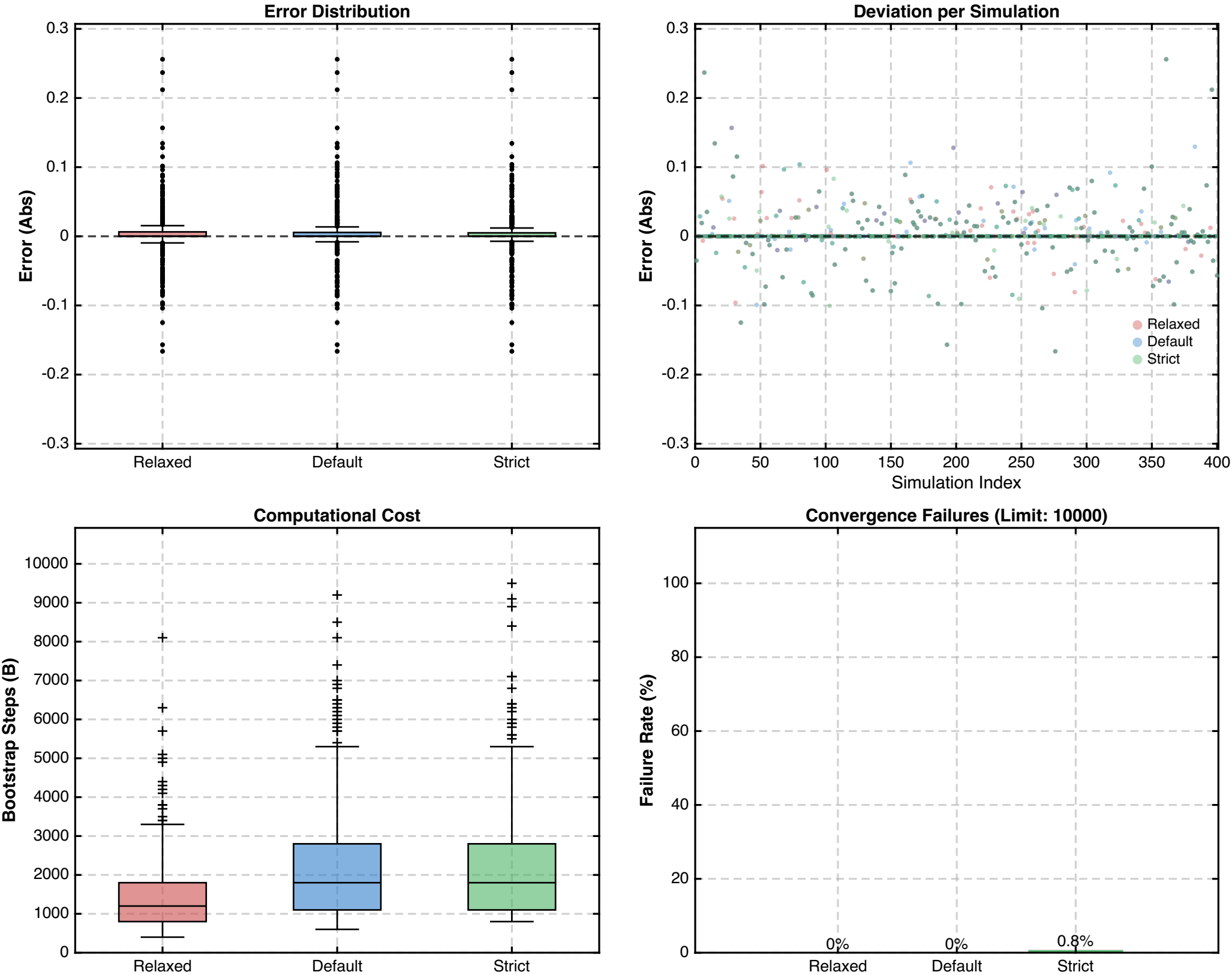
	Scenario	n	Distribution	Data Summary
1	**Baseline-S**	25	Normal	6 Datasets, Means 10.0-13.8, SD = 2.0
2	**Baseline-M**	50	Normal	6 Datasets, Means 10.0-13.8, SD = 2.0
3	**Baseline-L**	100	Normal	6 Datasets, Means 10.0-13.8, SD = 2.0
4	**Skewness**	50	Skewed	6 Datasets, Means 2.0-2.5 (Log), SD = 0.4 (Log)
5	**Quantization**	50	Likert	6 Datasets, Scale 1-7, Means 3.0-5.0, SD = 1.5
6	**Bimodality**	50	Bimodal	6 Datasets, Mix Means 10.0 & 15.0, Mode SD = 1.0
7	**Sensitivity**	50	Small Effect	6 Datasets, Means 10.0-11.3, SD = 2.0
8	**Discrimination**	50	Large Effect	6 Datasets, Means 10.0-17.5, SD = 2.0

Configuration: Method Parameters

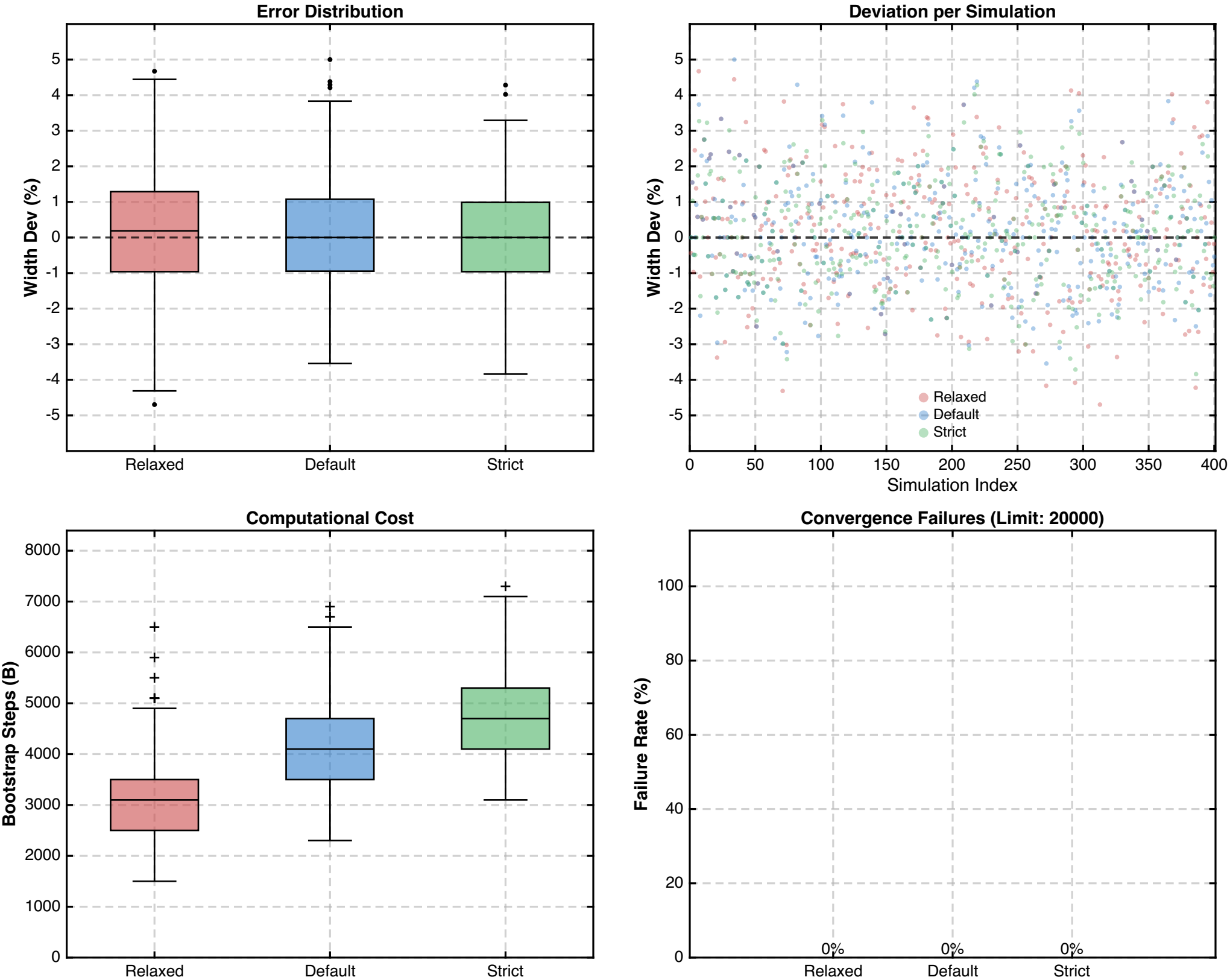
References (B): Thresholds = 25000 | BCa = 50000 | Ranking = 10000

Parameter Set	Relaxed	Default	Strict
1. Bootstrap Thresholds			
Trials (n)	15	25	35
Smoothing (sm)	2	3	4
Streak (st)	2	3	4
Tolerance (tol)	0.010	0.010	0.010
Step Size (B)	100	100	100
B Range	[100-10000]	[100-10000]	[100-10000]
2. BCa Confidence Intervals			
Trials (n)	20	30	40
Smoothing (sm)	2	3	4
Streak (st)	2	3	4
Tolerance (tol)	0.030	0.030	0.030
Step Size (B)	200	200	200
B Range	[100-20000]	[100-20000]	[100-20000]
3. Ranking Stability			
Trials (n)	10	15	20
Smoothing (sm)	2	3	4
Streak (st)	2	3	4
Tolerance (tol)	0.005	0.005	0.005
Step Size (B)	25	25	25
B Range	[50-2500]	[50-2500]	[50-2500]

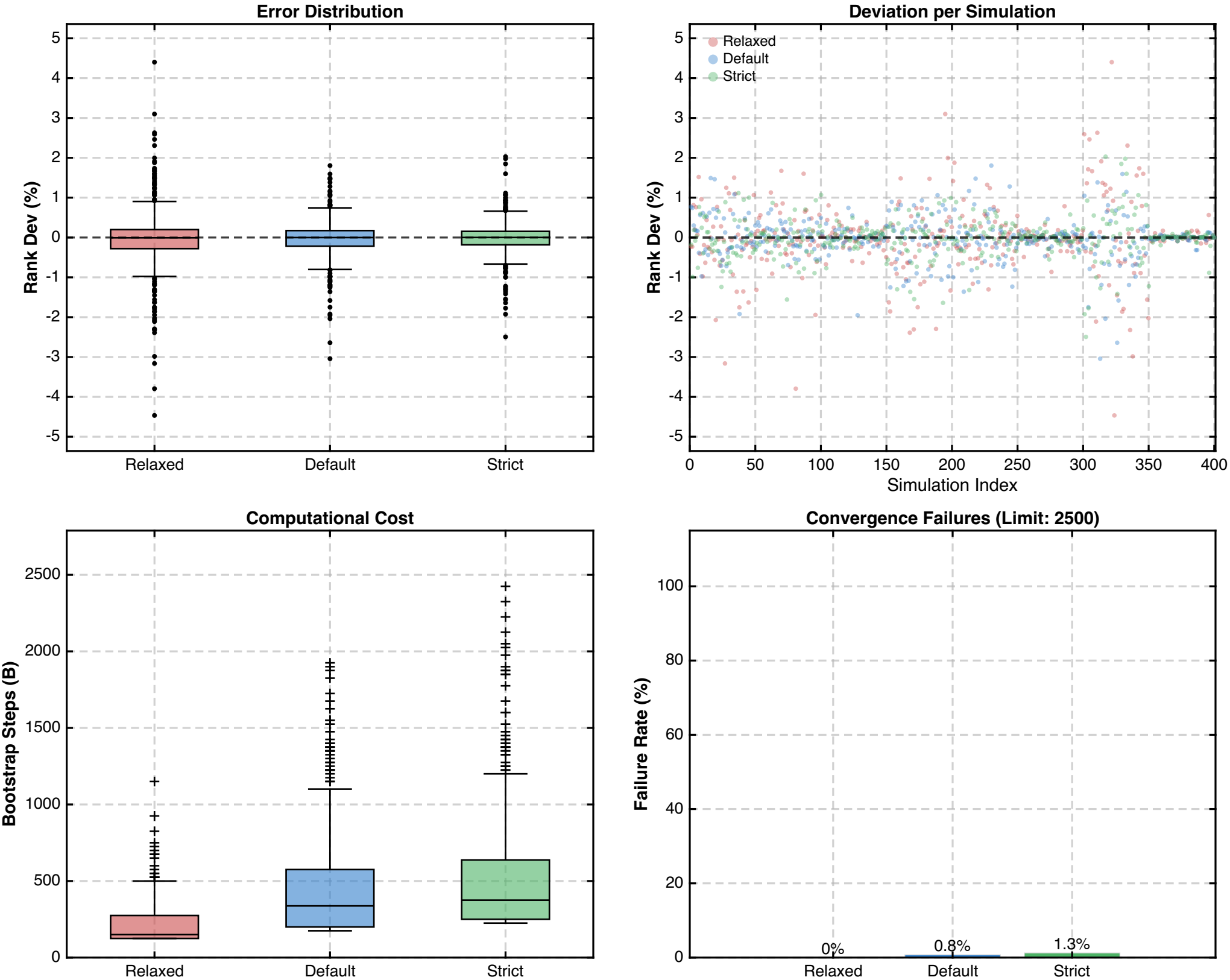
Thresholds (Delta) Analysis: Global Summary (Ref B = 25000)



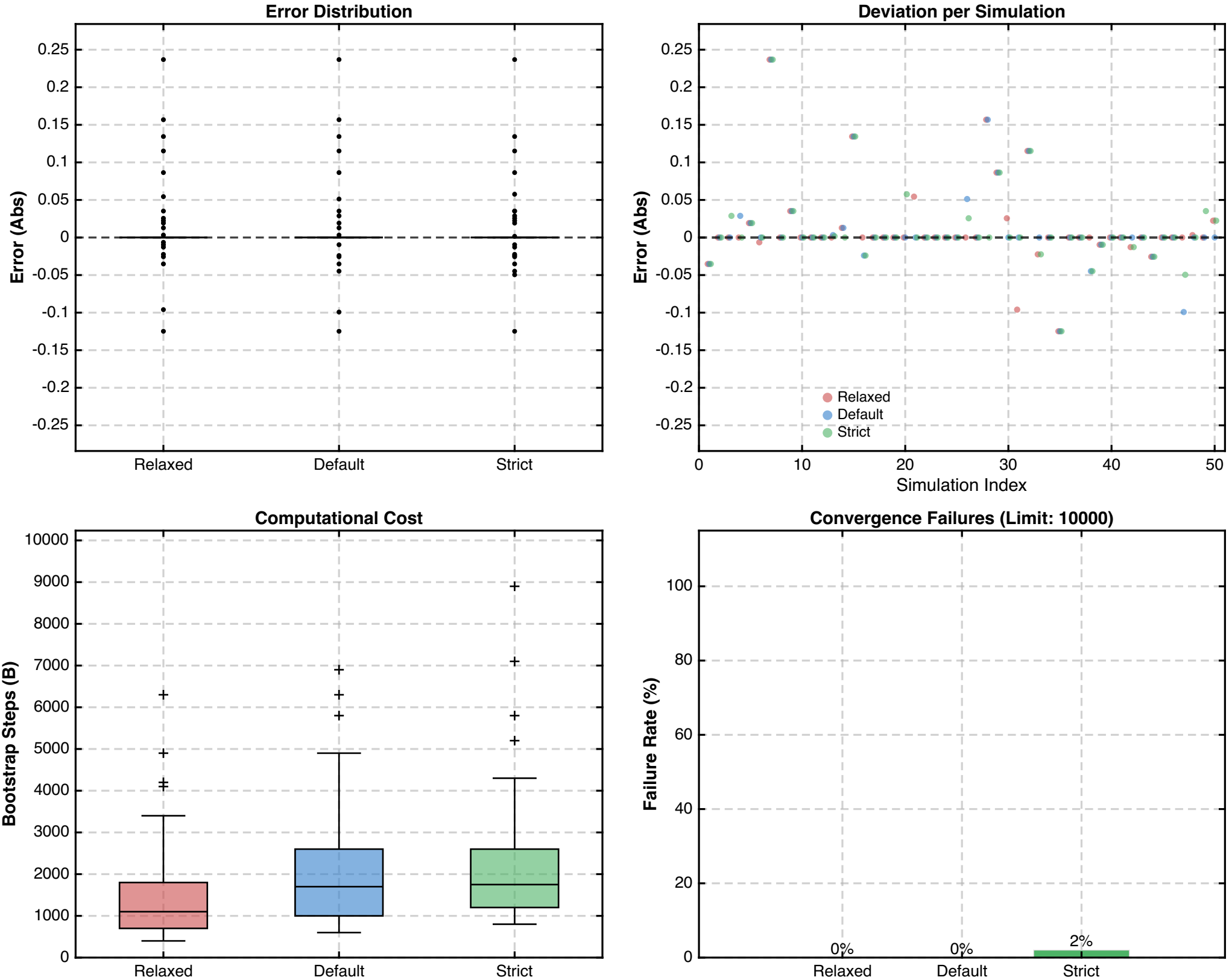
BCa CI (Width) Analysis: Global Summary (Ref B = 50000)



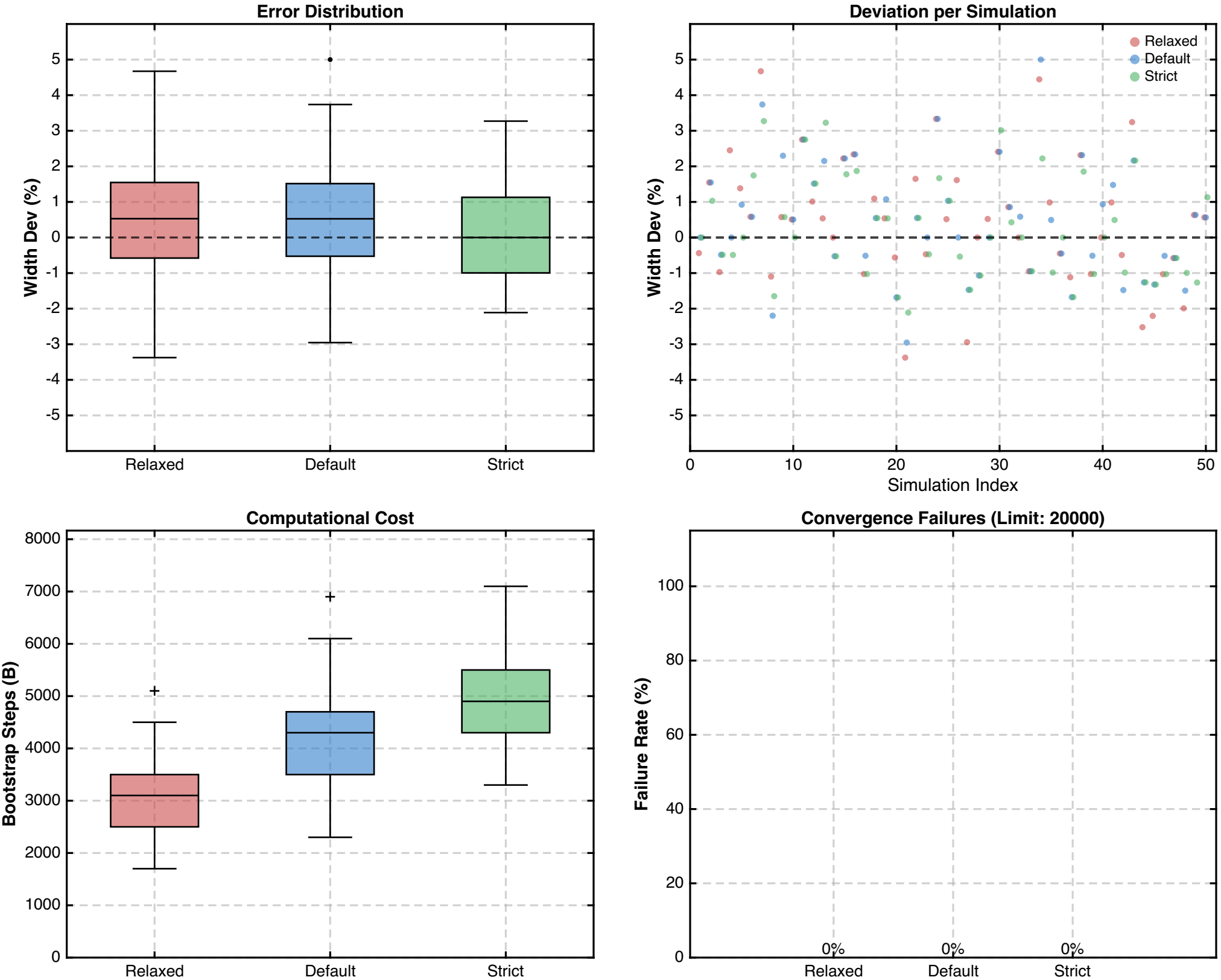
Ranking (Mean) Analysis: Global Summary (Ref B = 10000)



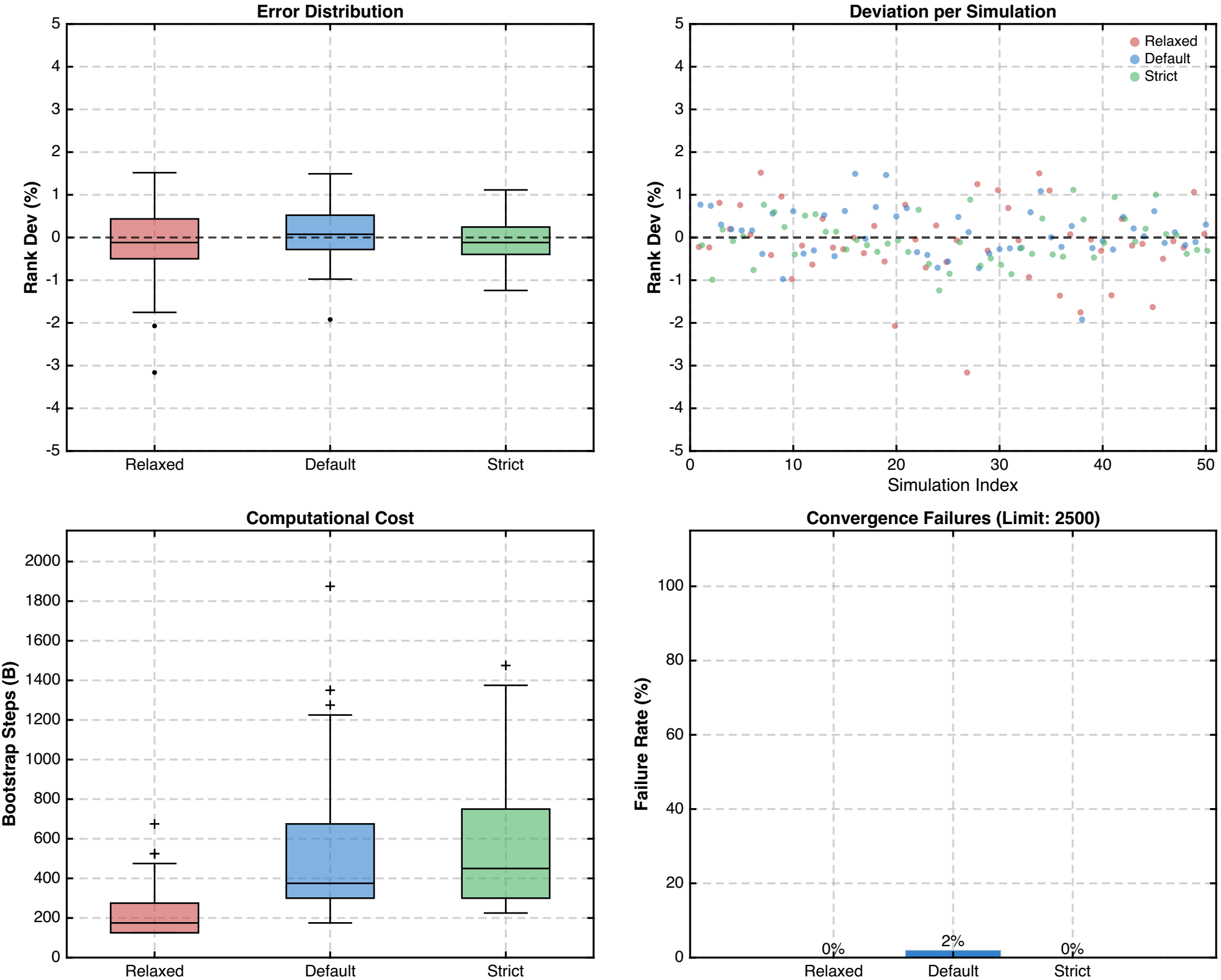
Thresholds (Delta) Analysis: Baseline S (Median Cliff's d = 0.45)



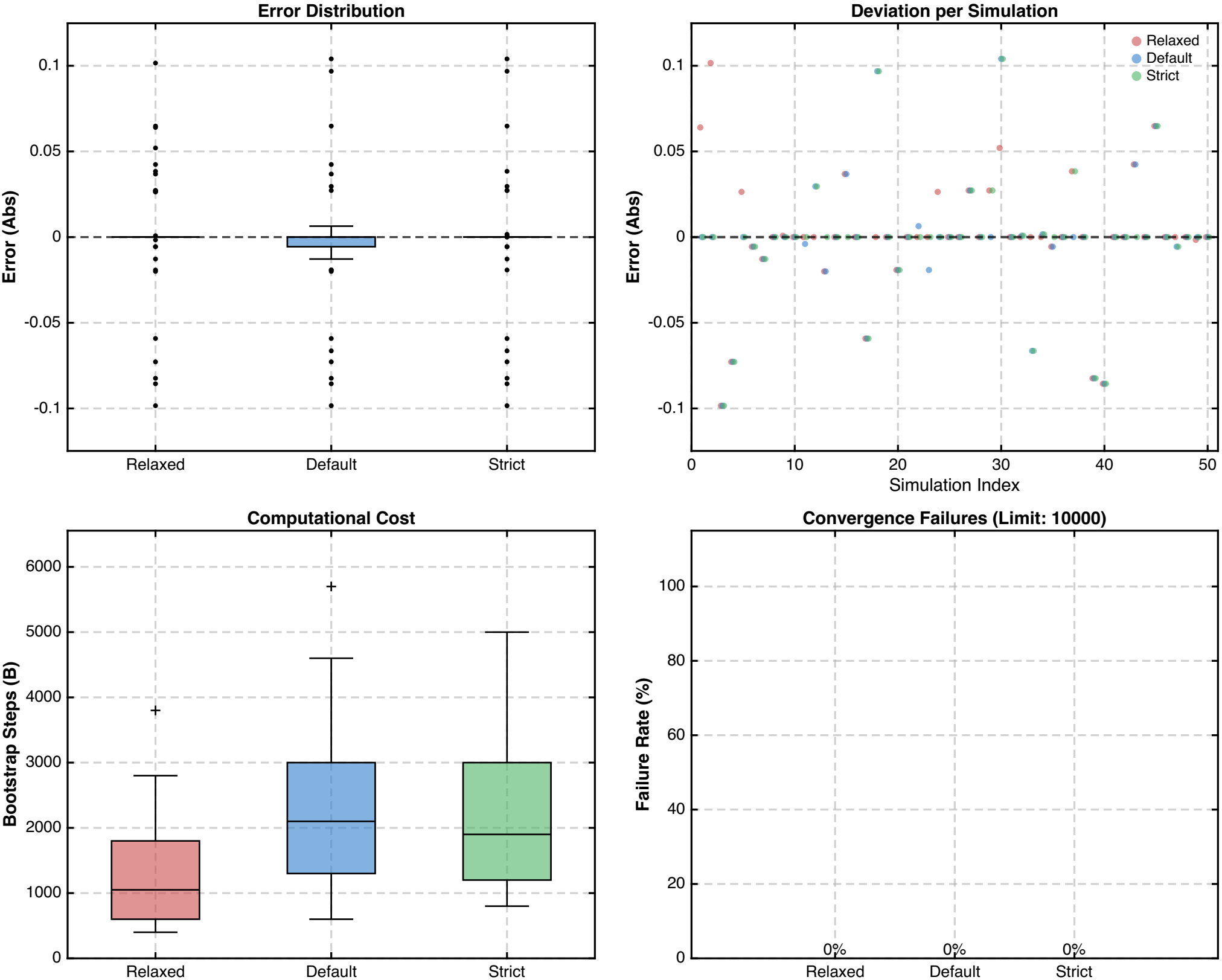
BCa CI (Width) Analysis: Baseline S (Median Cliff's d = 0.45)



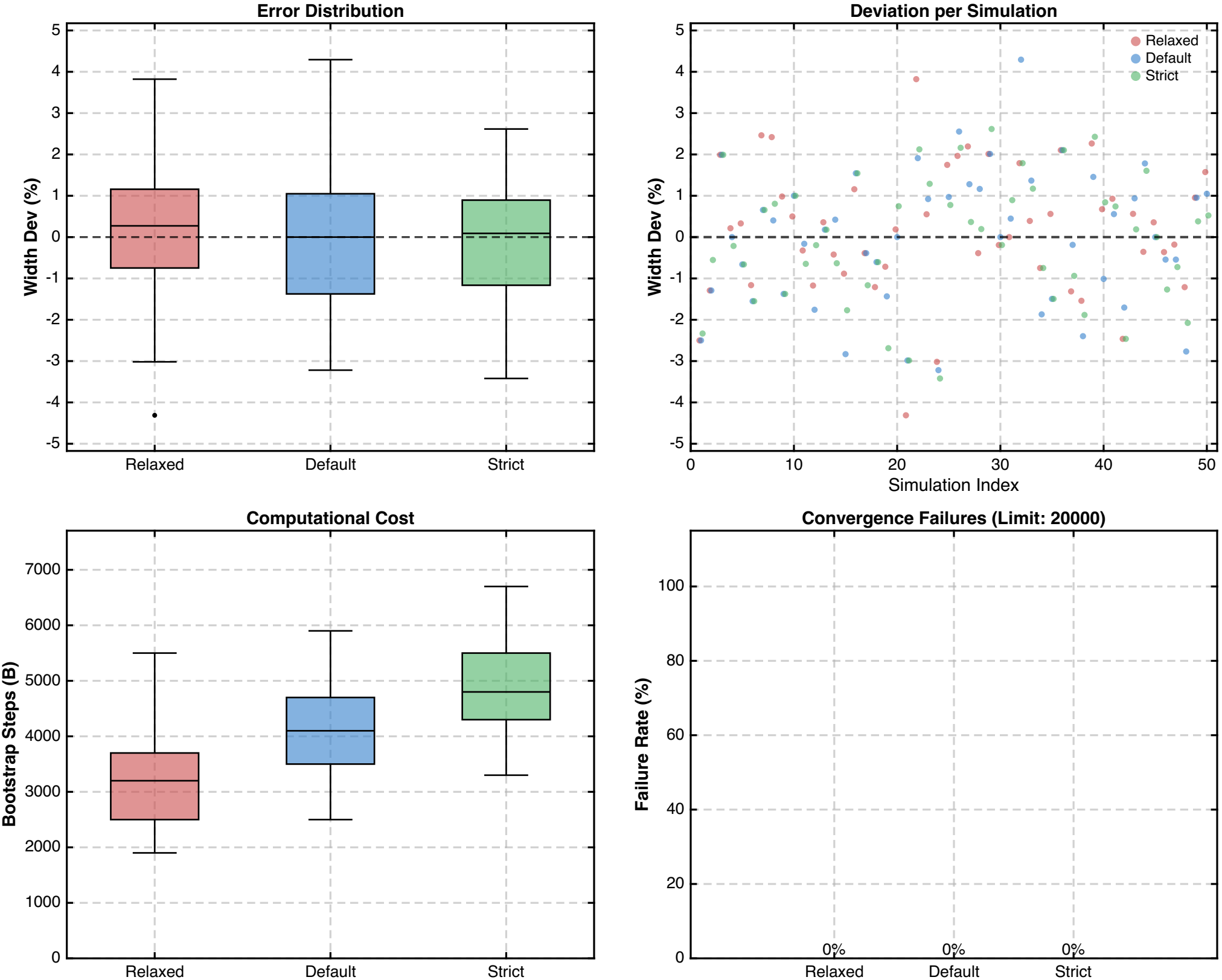
Ranking (Mean) Analysis: Baseline S (Median Cliff's d = 0.45)



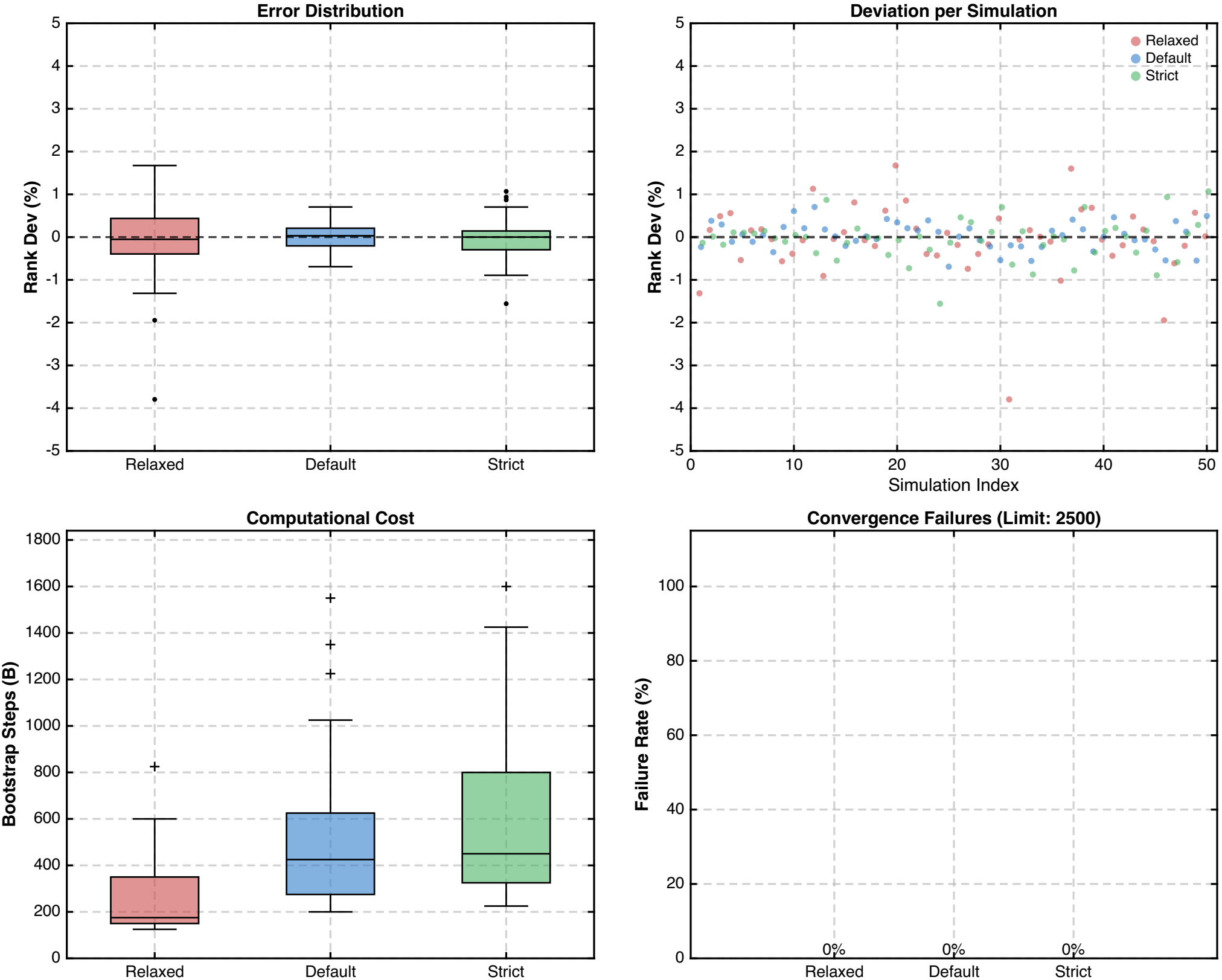
Thresholds (Delta) Analysis: Baseline M (Median Cliff's d = 0.42)



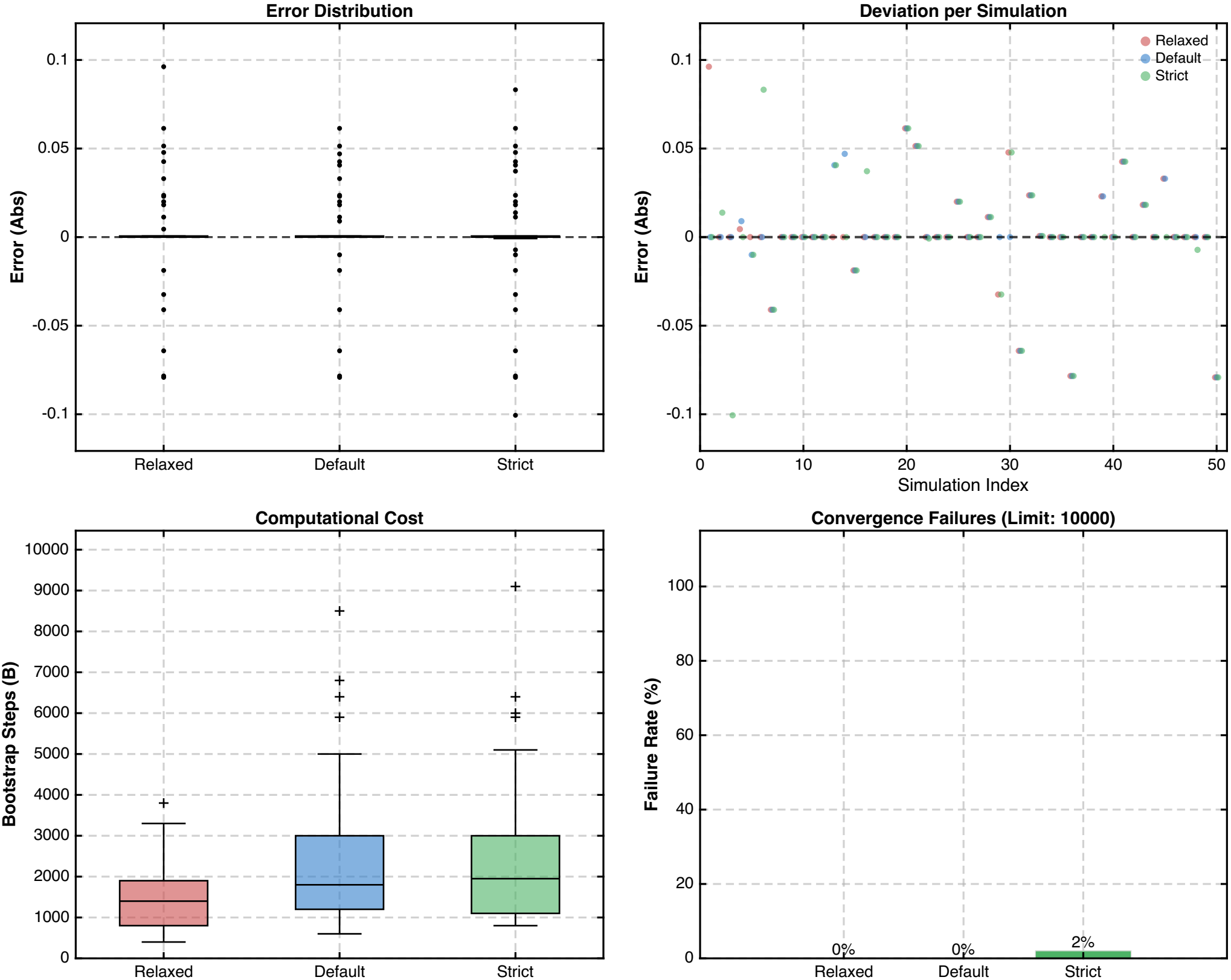
BCa CI (Width) Analysis: Baseline M (Median Cliff's d = 0.42)



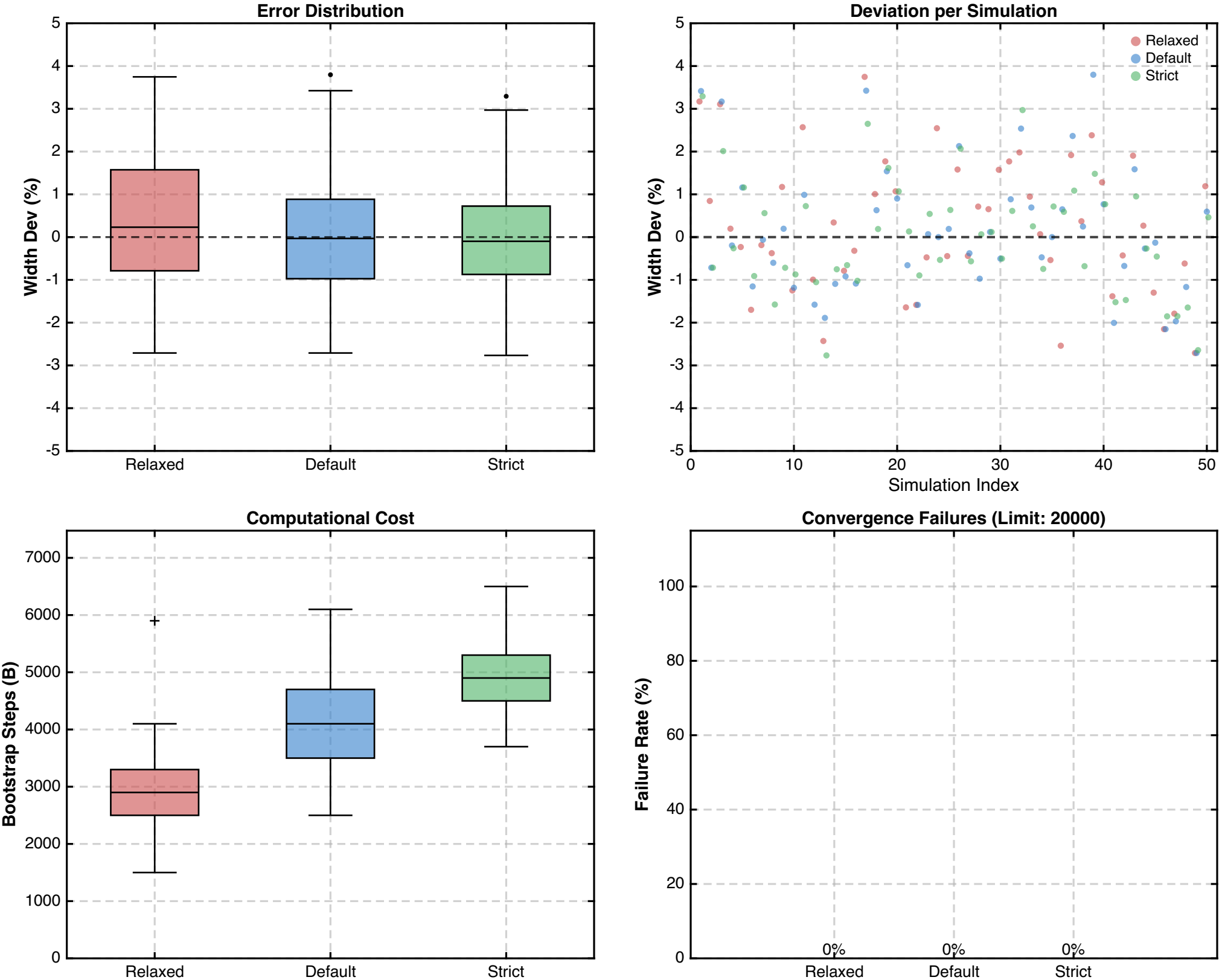
Ranking (Mean) Analysis: Baseline M (Median Cliff's d = 0.42)



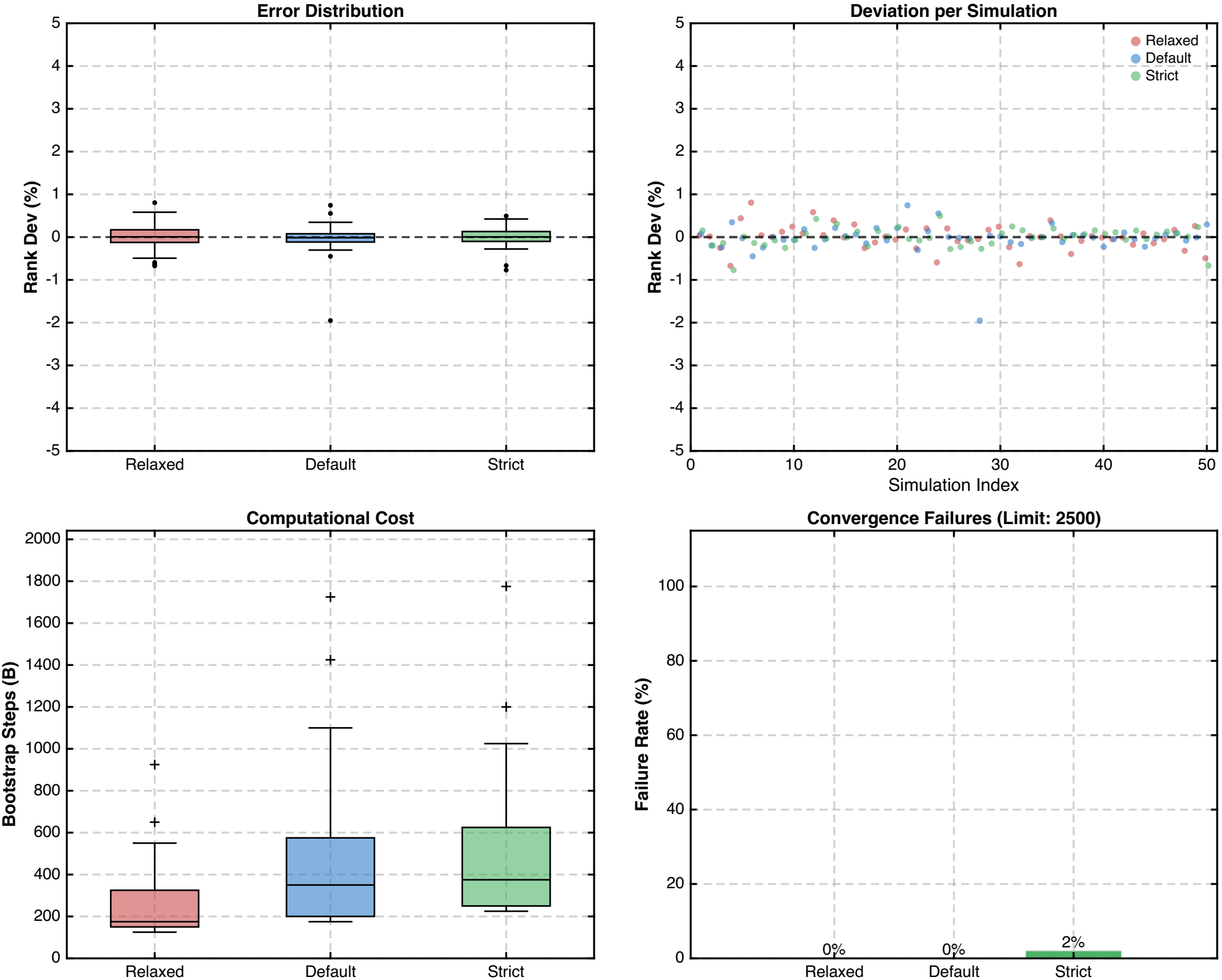
Thresholds (Delta) Analysis: Baseline L (Median Cliff's d = 0.42)



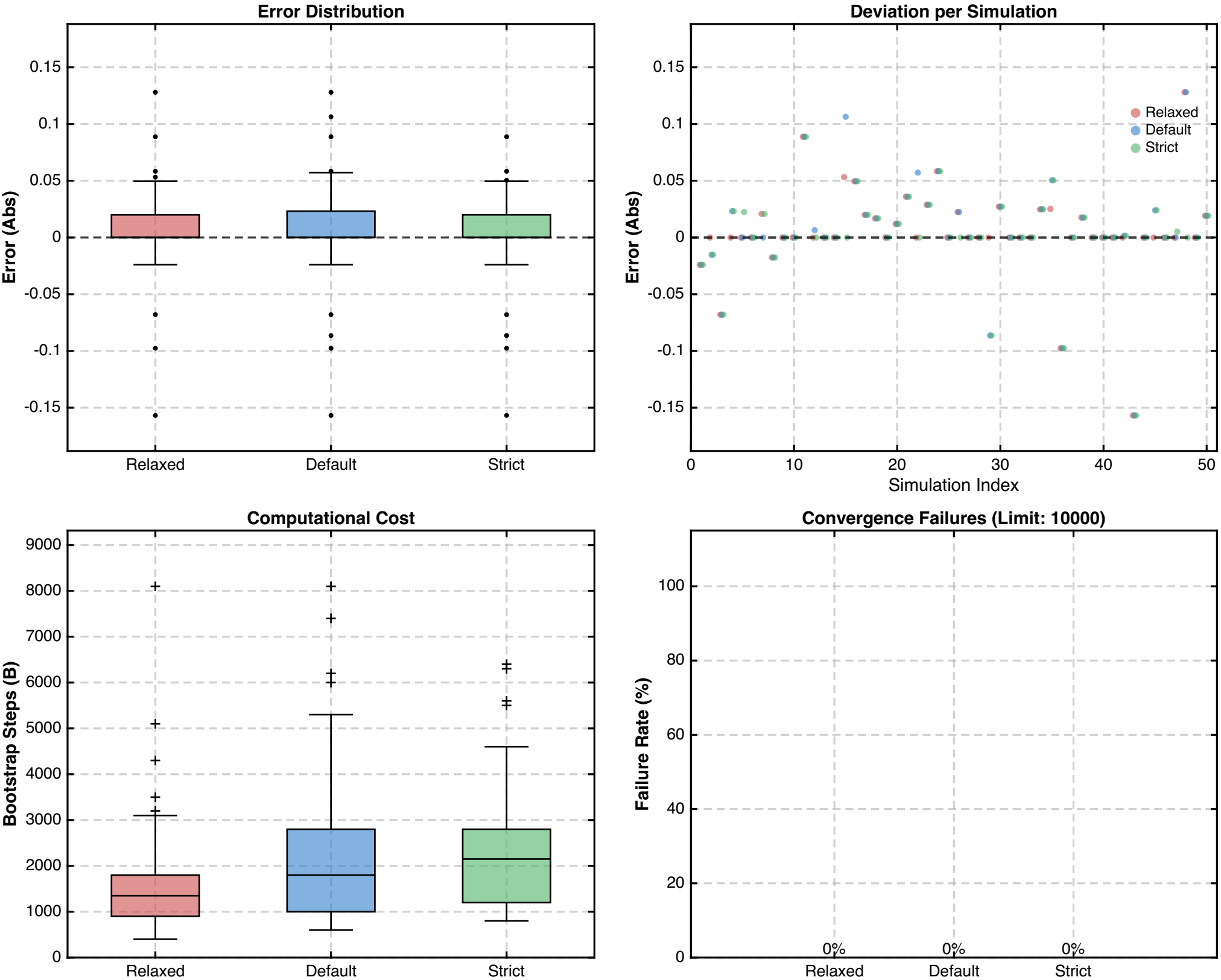
BCa CI (Width) Analysis: Baseline L (Median Cliff's d = 0.42)



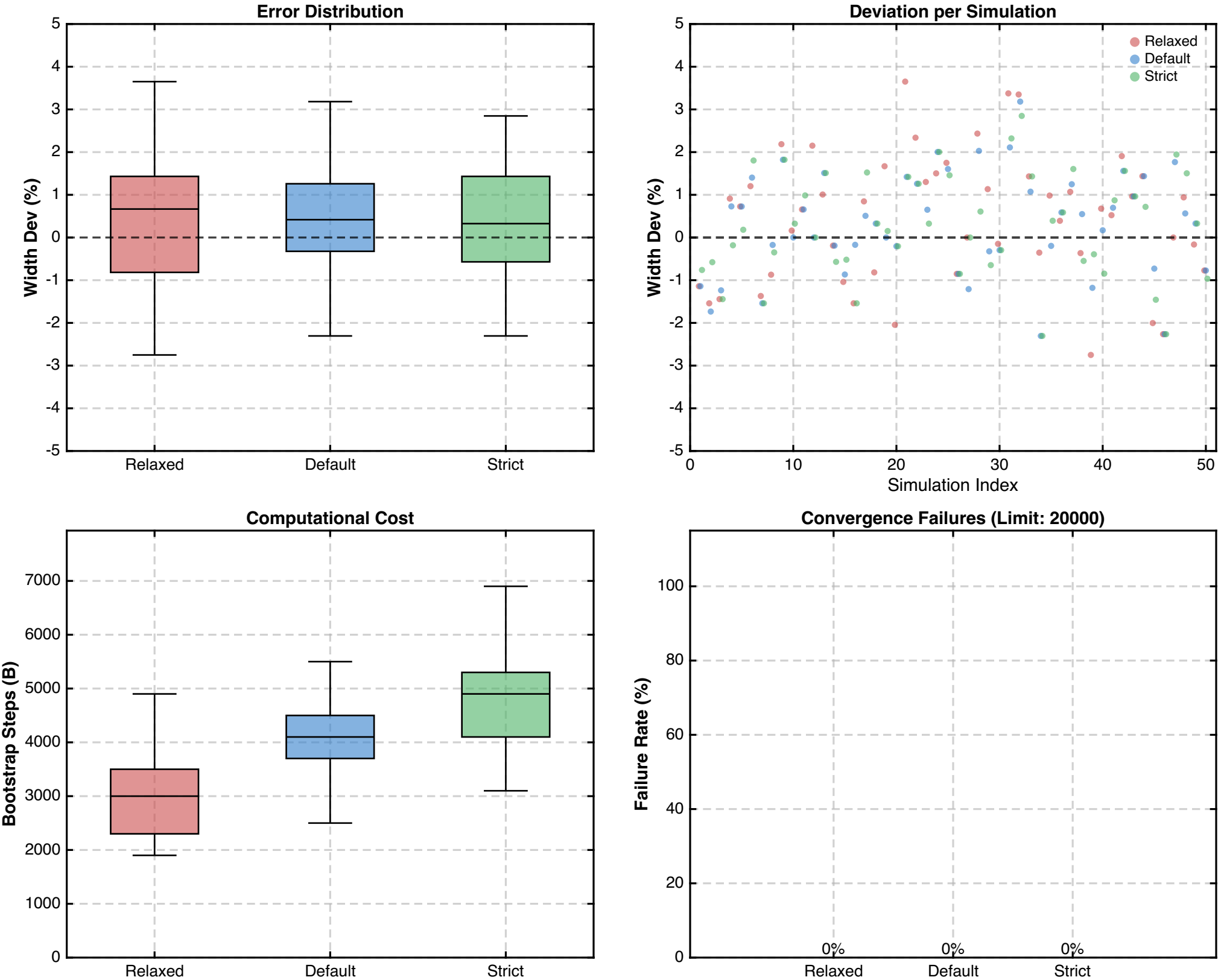
Ranking (Mean) Analysis: Baseline L (Median Cliff's d = 0.42)



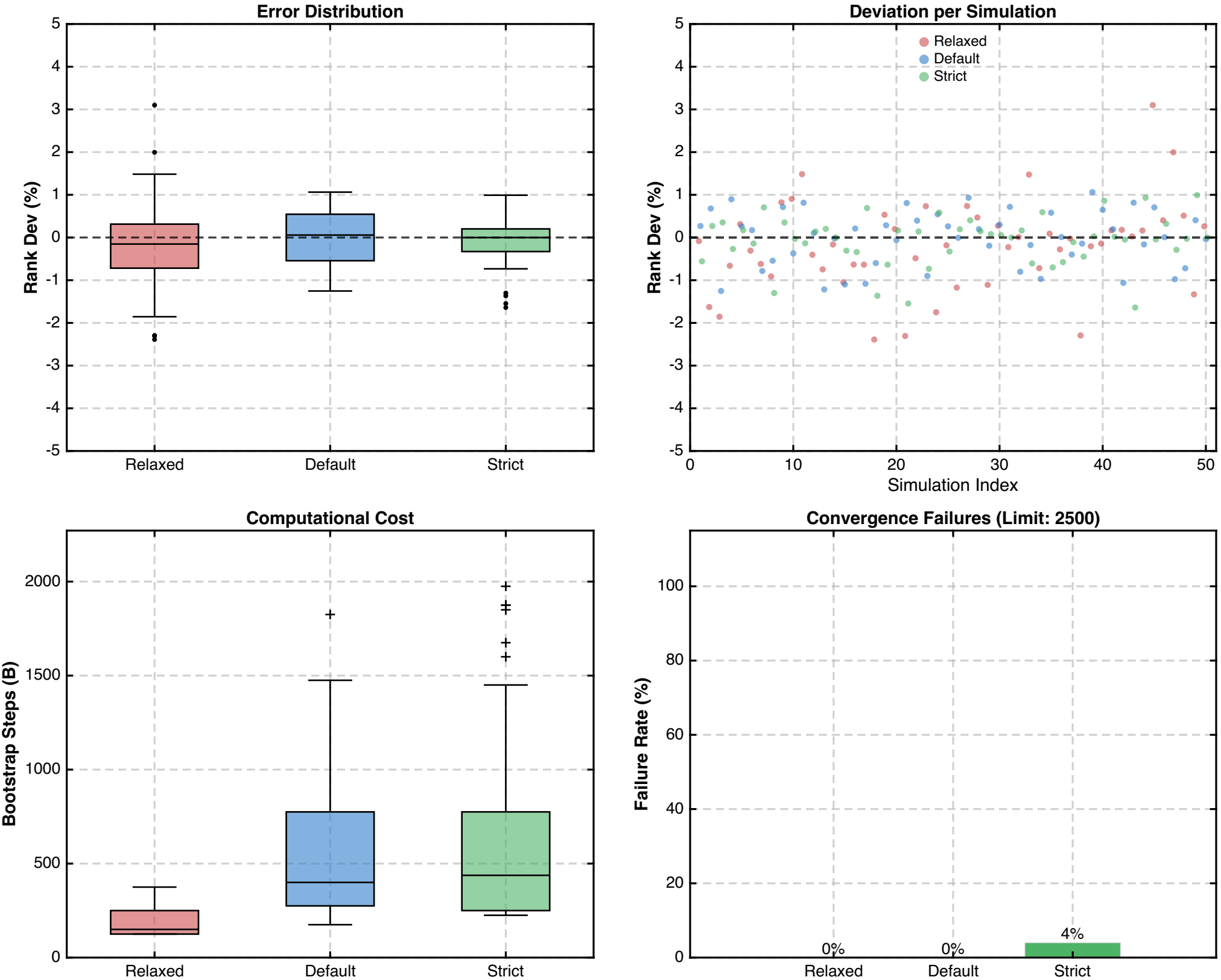
Thresholds (Delta) Analysis: Skewness (Median Cliff's d = 0.29)



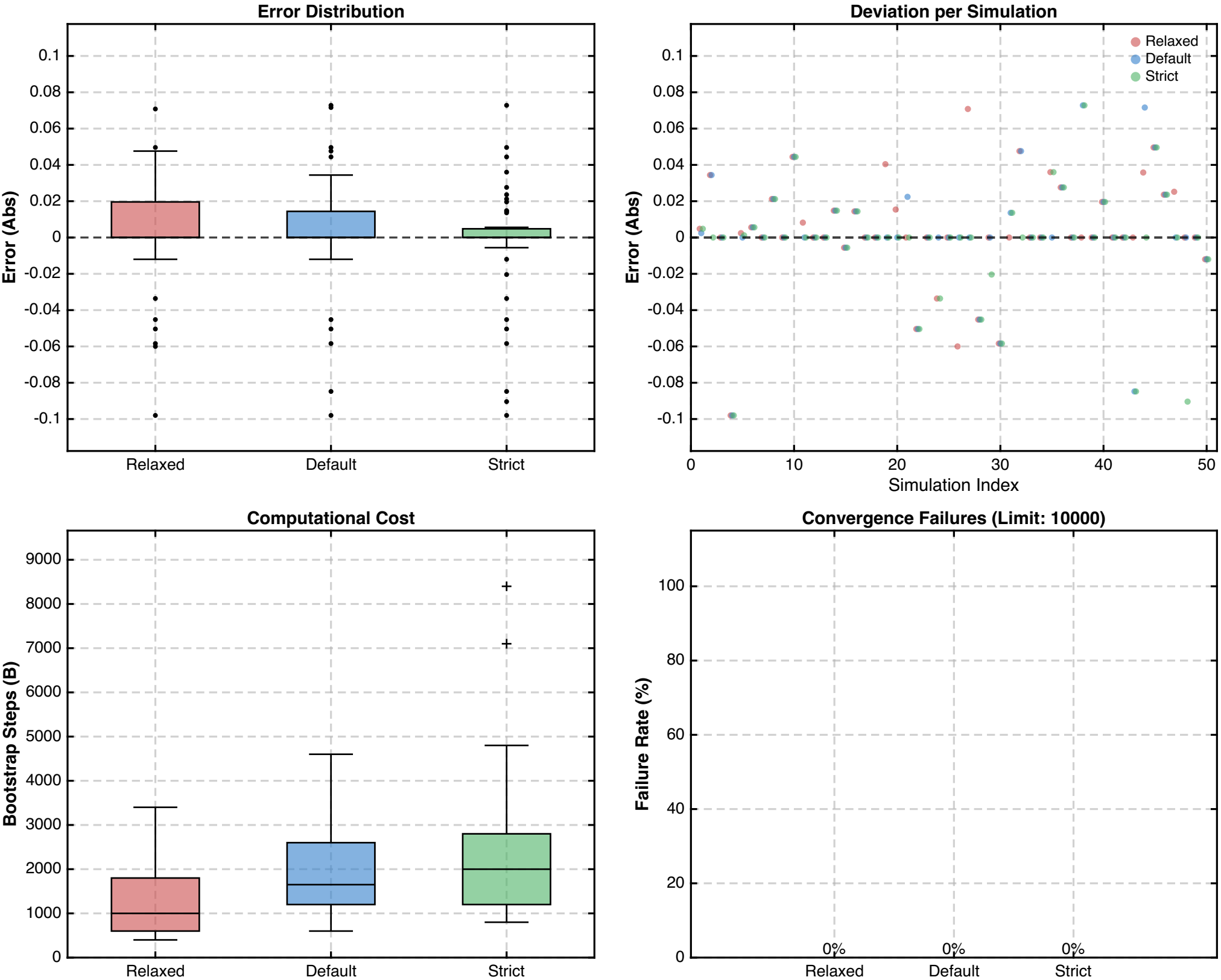
BCa CI (Width) Analysis: Skewness (Median Cliff's d = 0.29)



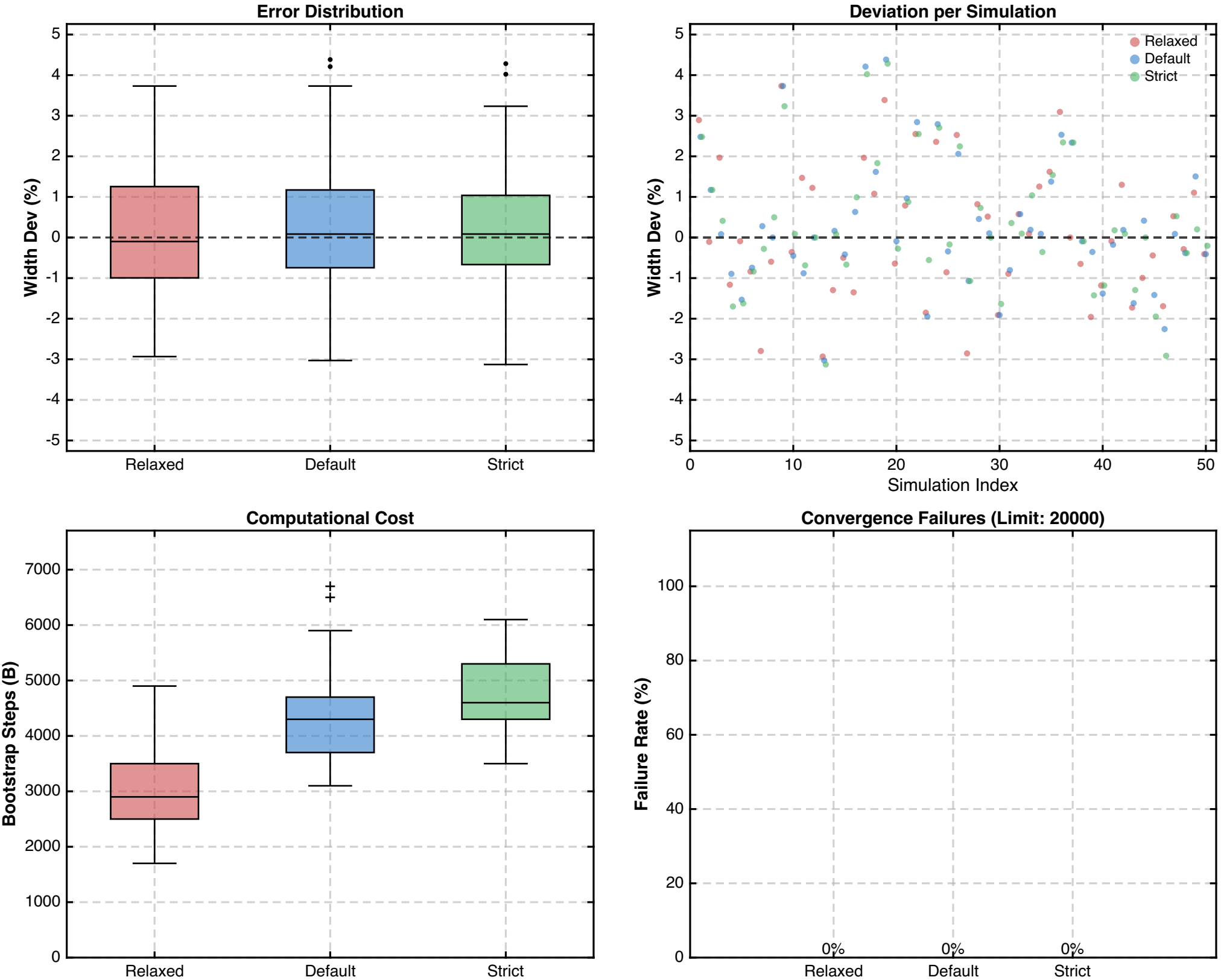
Ranking (Mean) Analysis: Skewness (Median Cliff's d = 0.29)



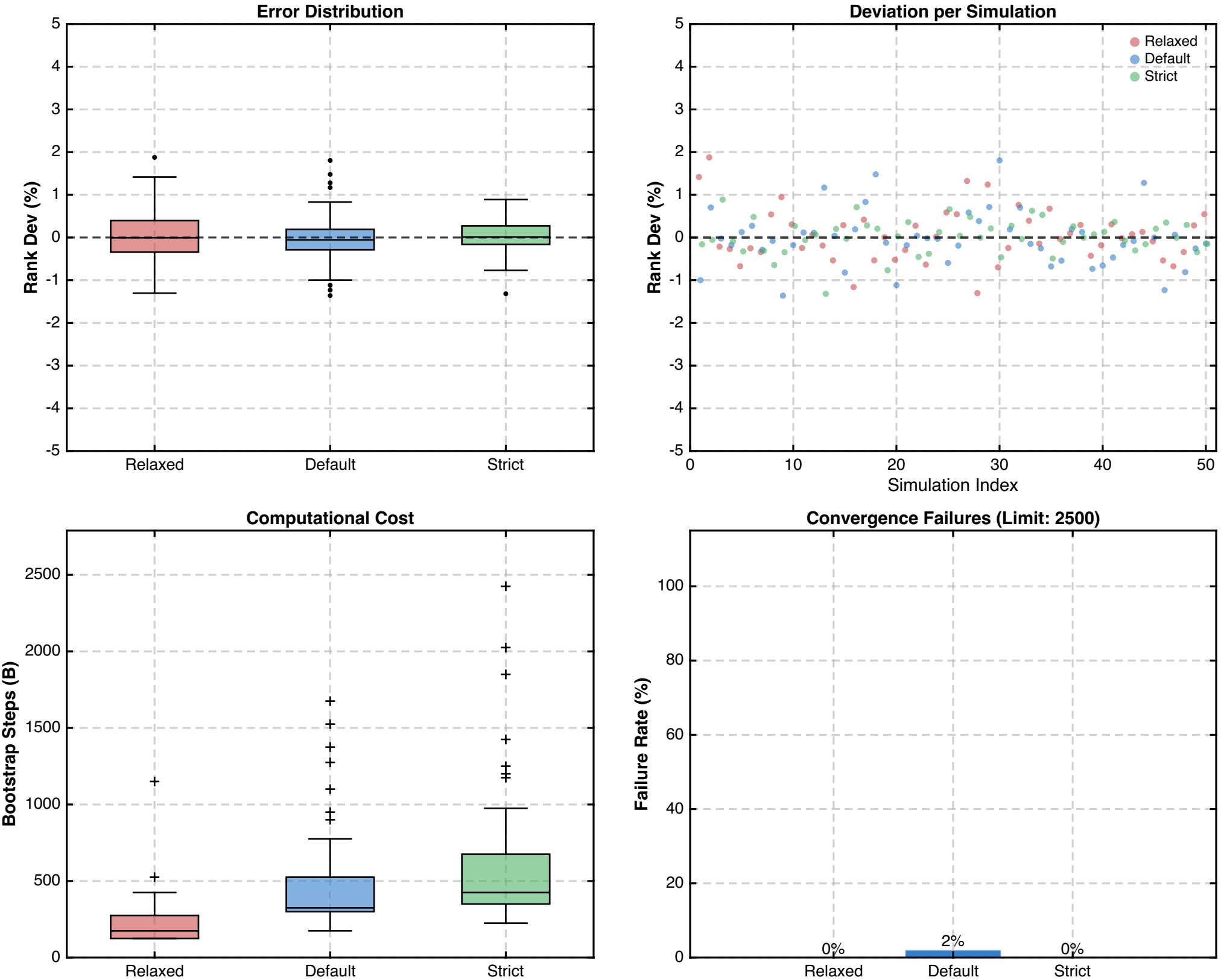
Thresholds (Delta) Analysis: Quantization (Median Cliff's d = 0.31)



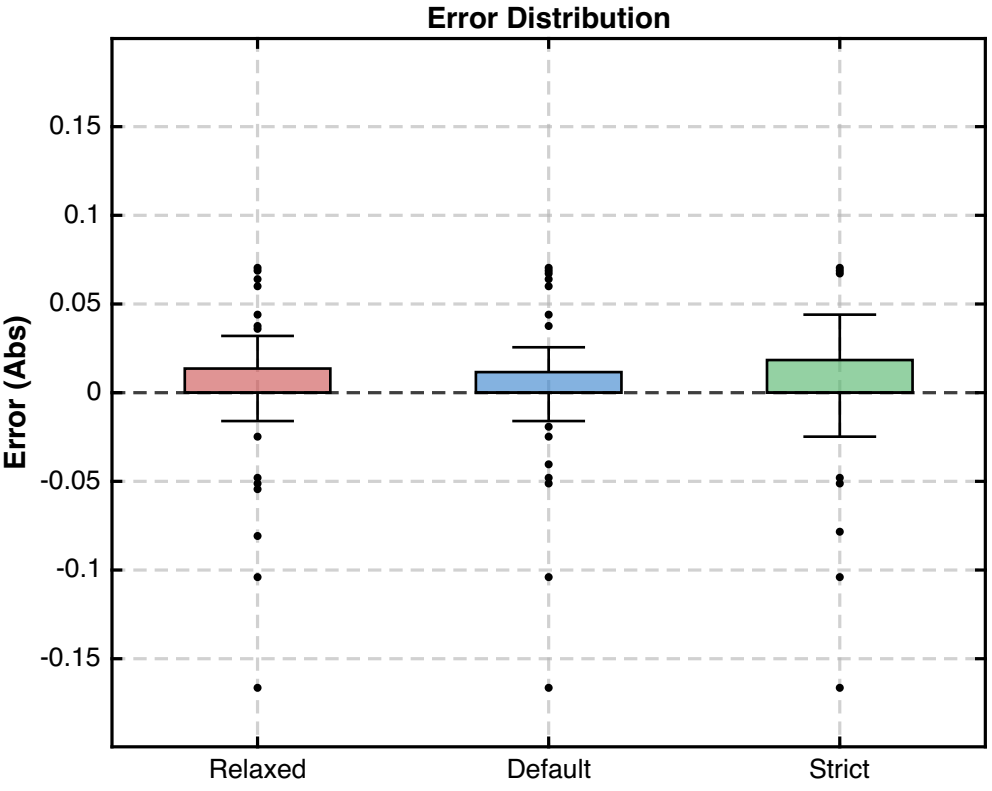
BCa CI (Width) Analysis: Quantization (Median Cliff's d = 0.31)

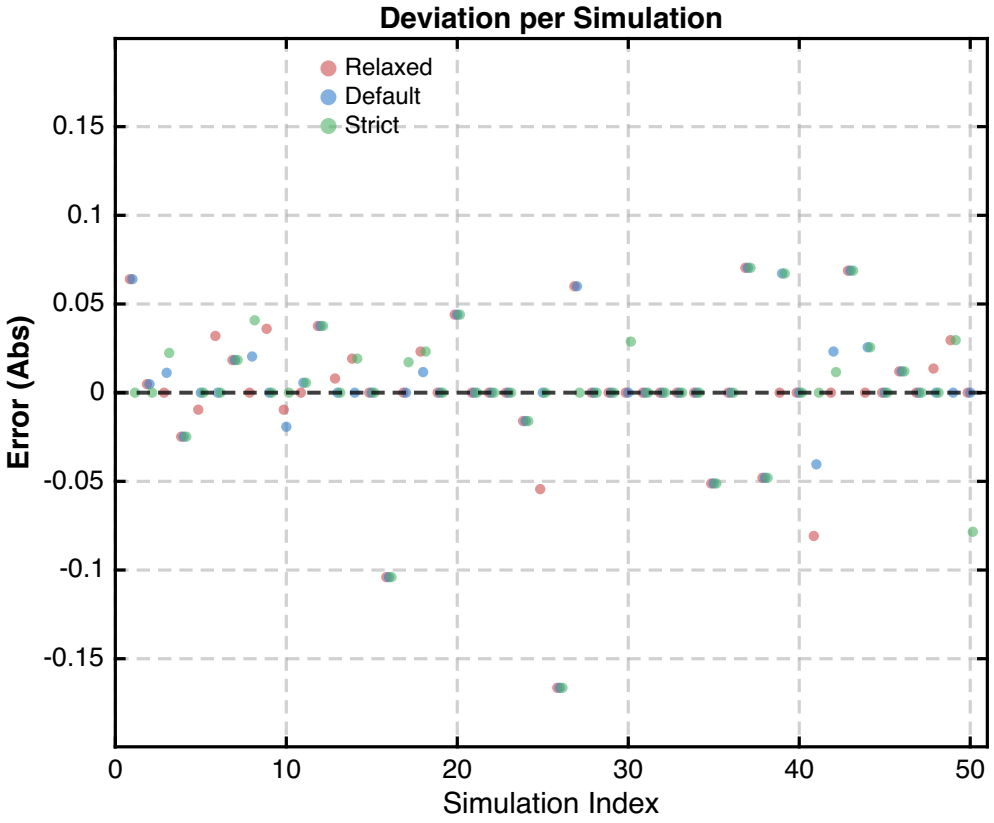


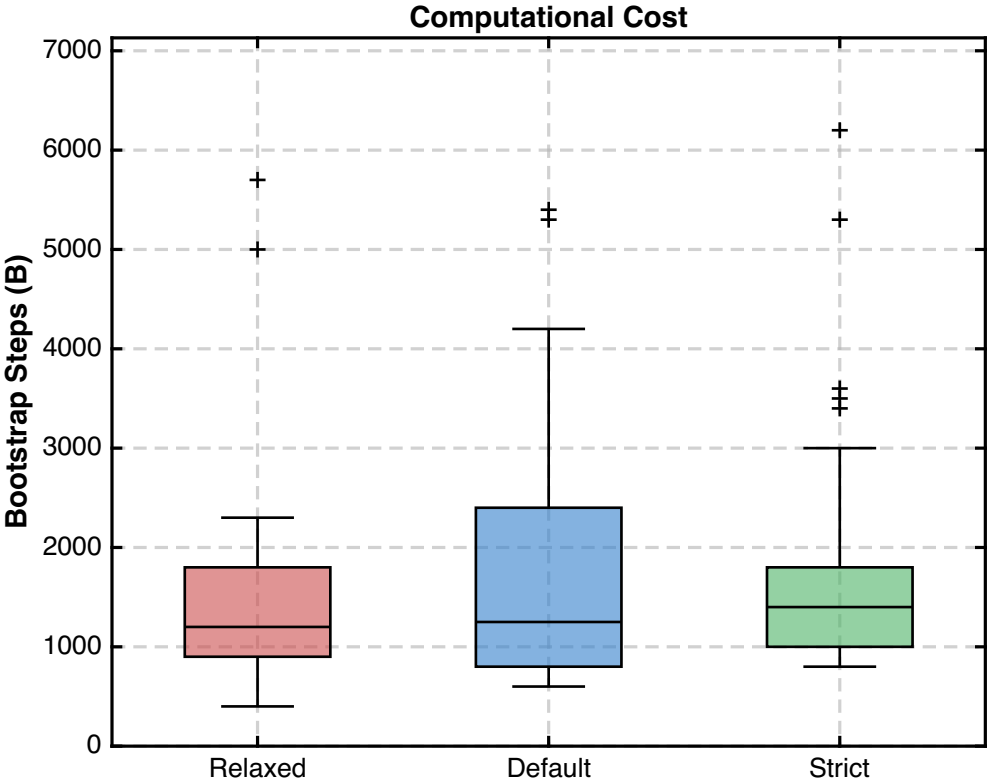
Ranking (Mean) Analysis: Quantization (Median Cliff's d = 0.31)

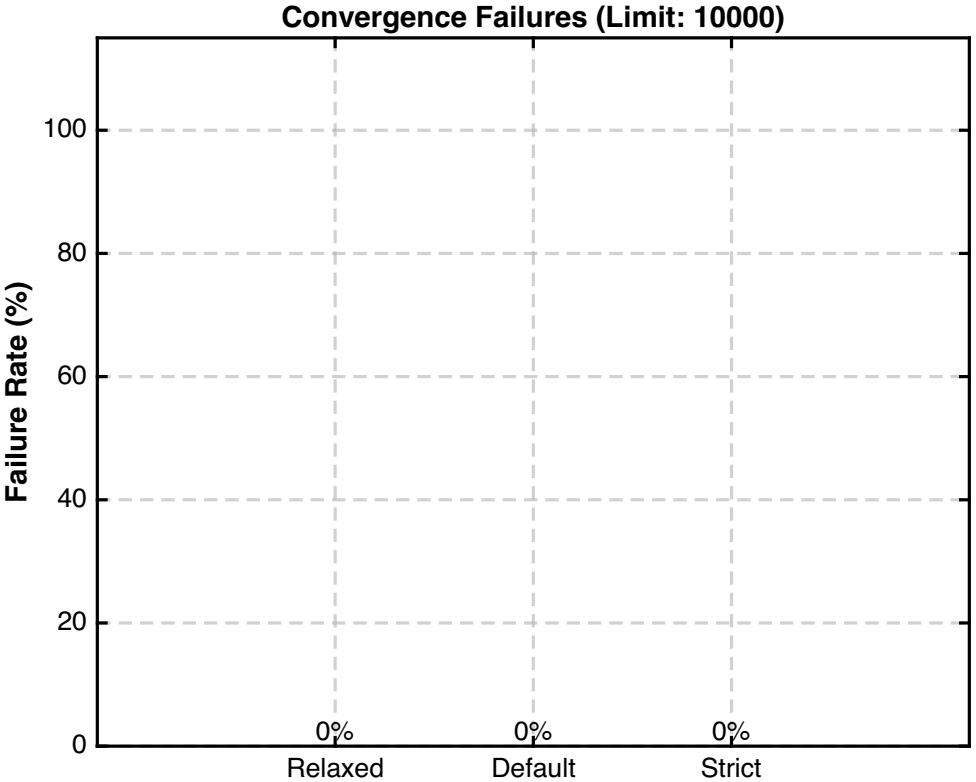


Thresholds (Delta) Analysis: Bimodality (Median Cliff's d = 0.56)

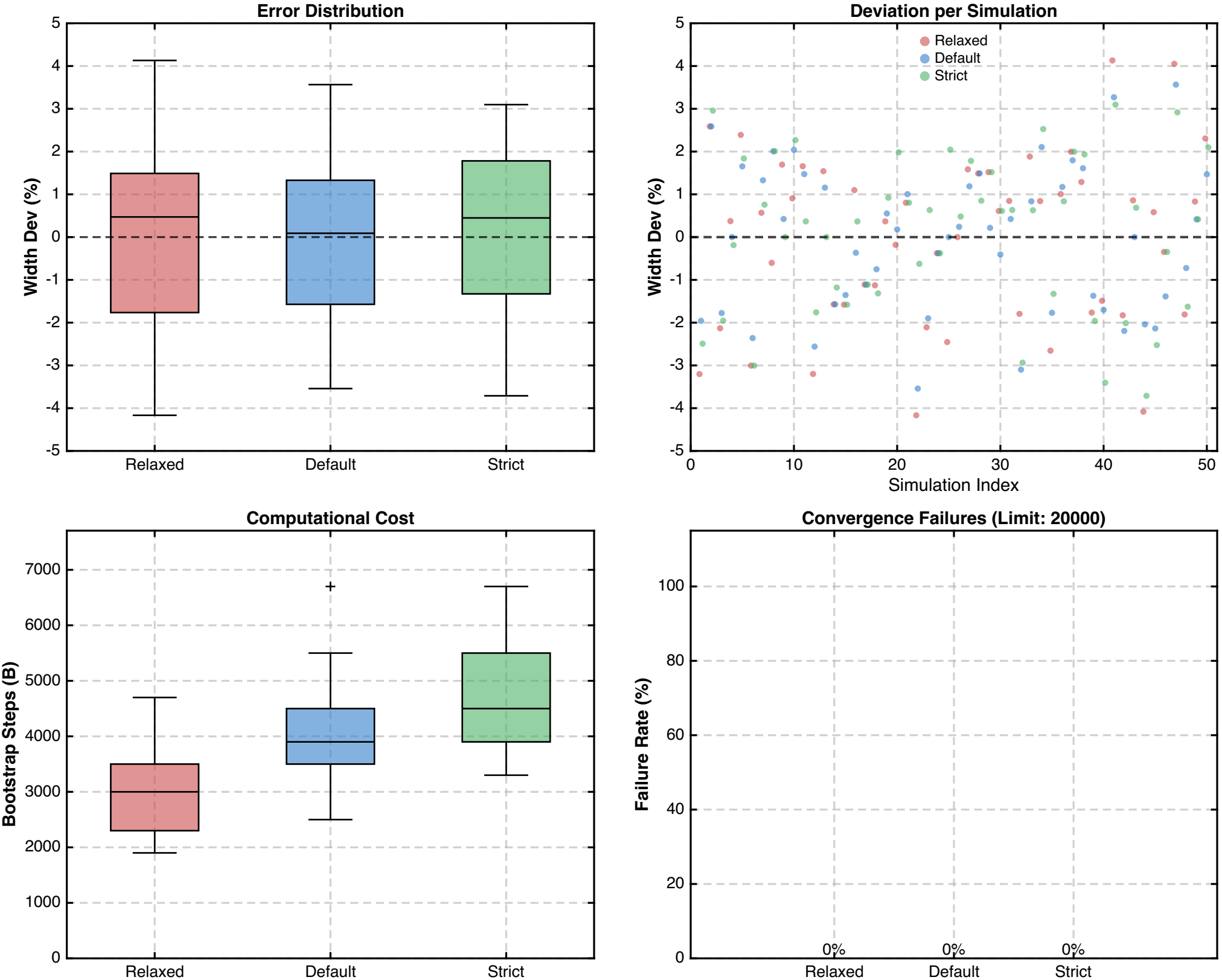




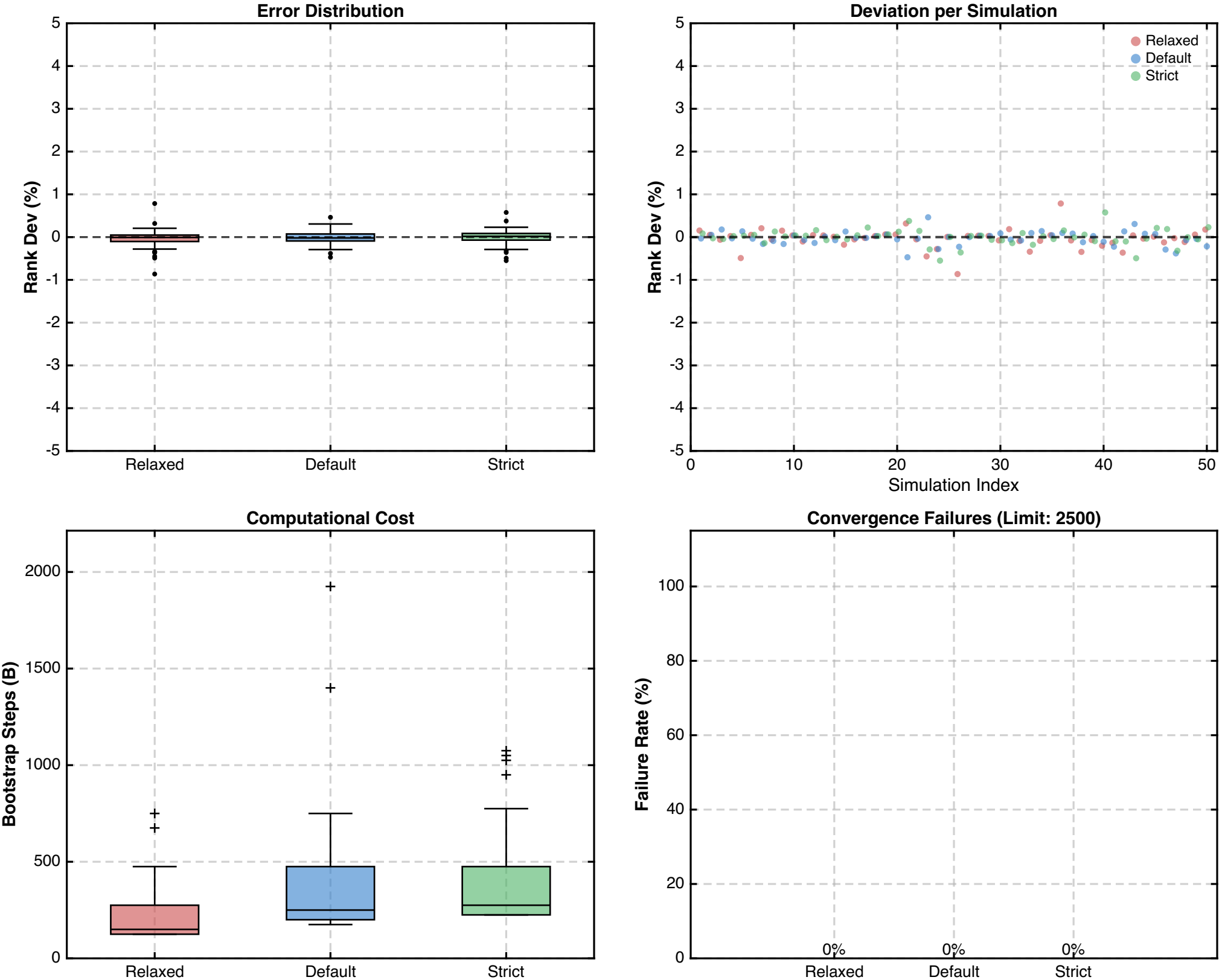




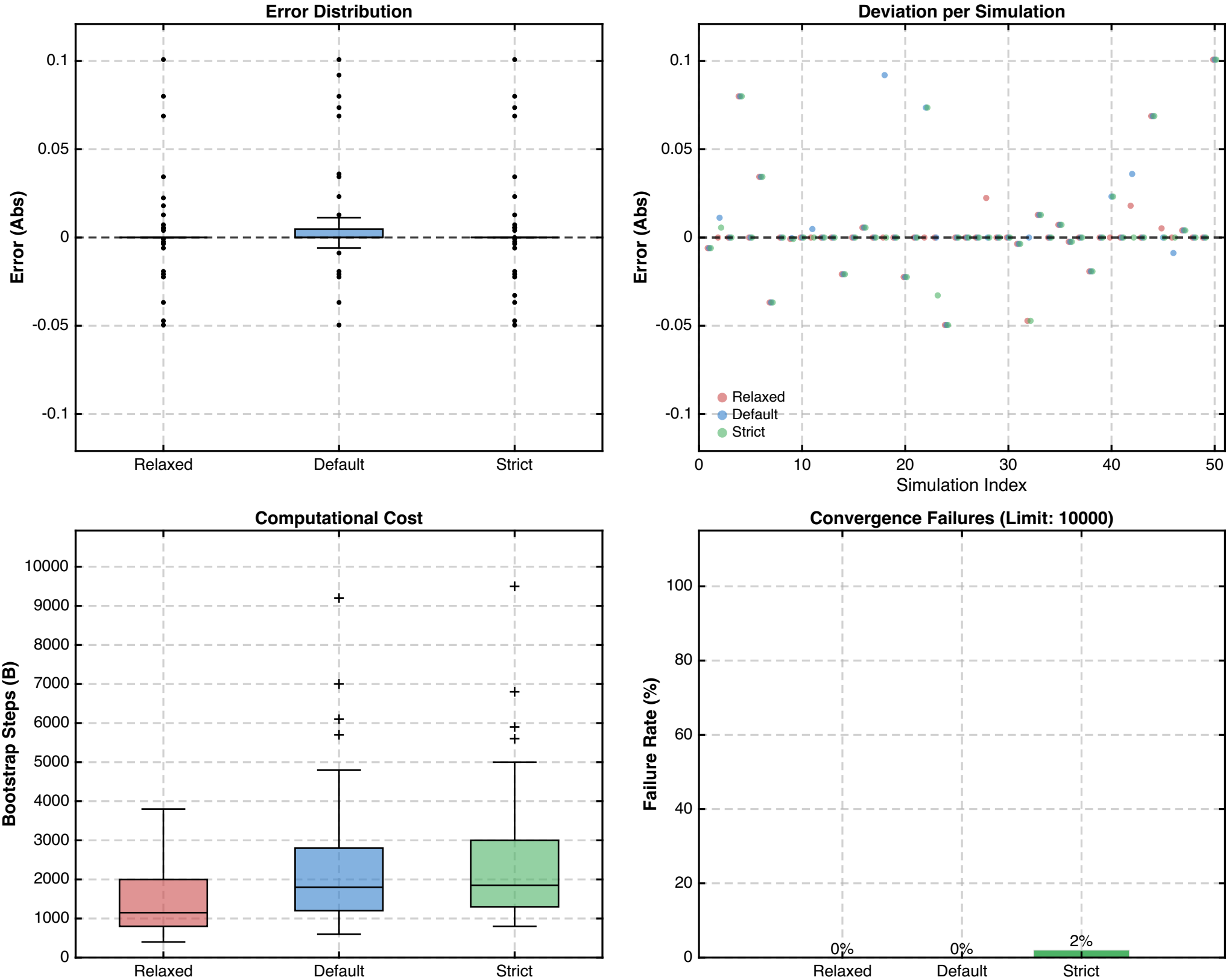
BCa CI (Width) Analysis: Bimodality (Median Cliff's d = 0.56)



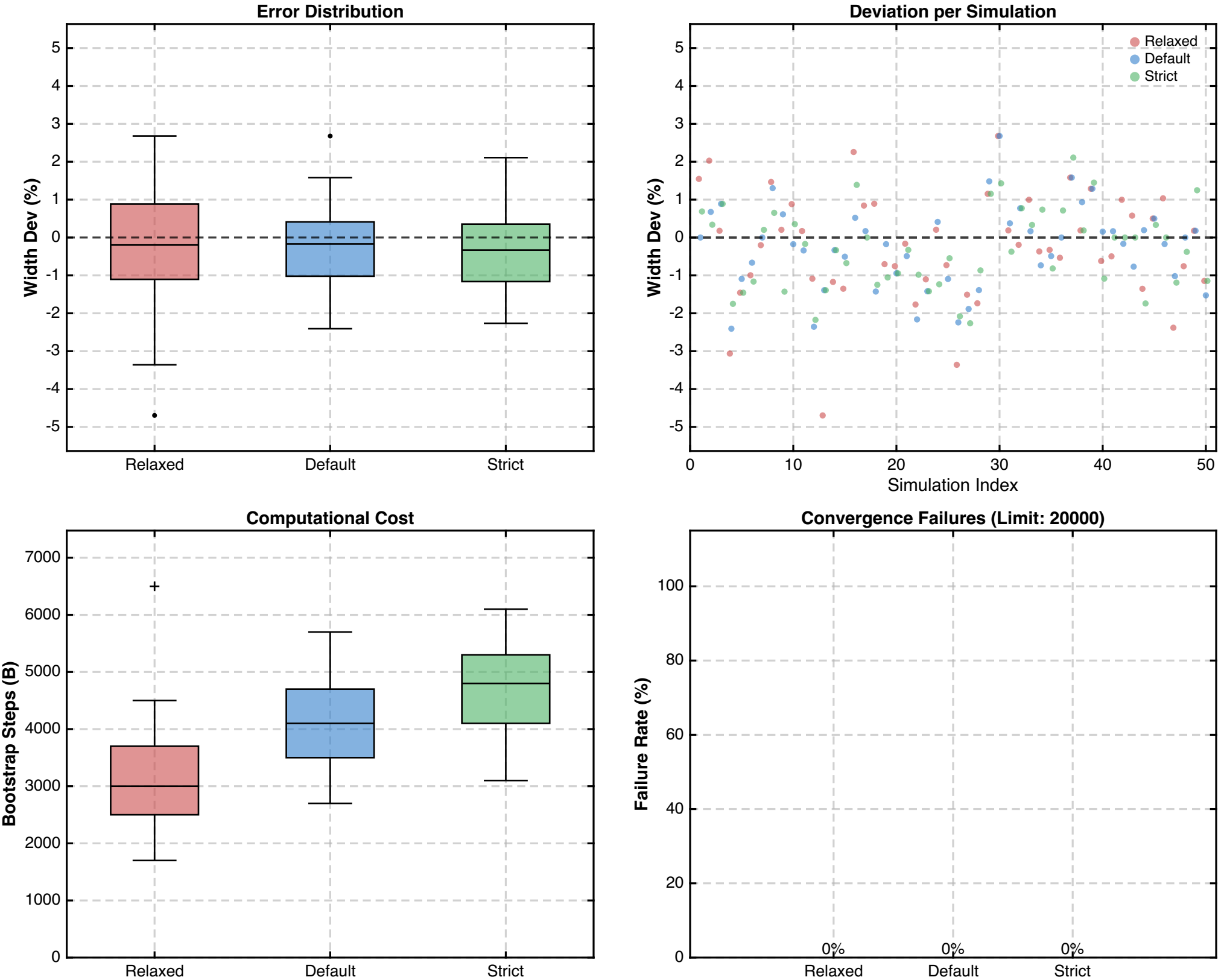
Ranking (Mean) Analysis: Bimodality (Median Cliff's d = 0.56)



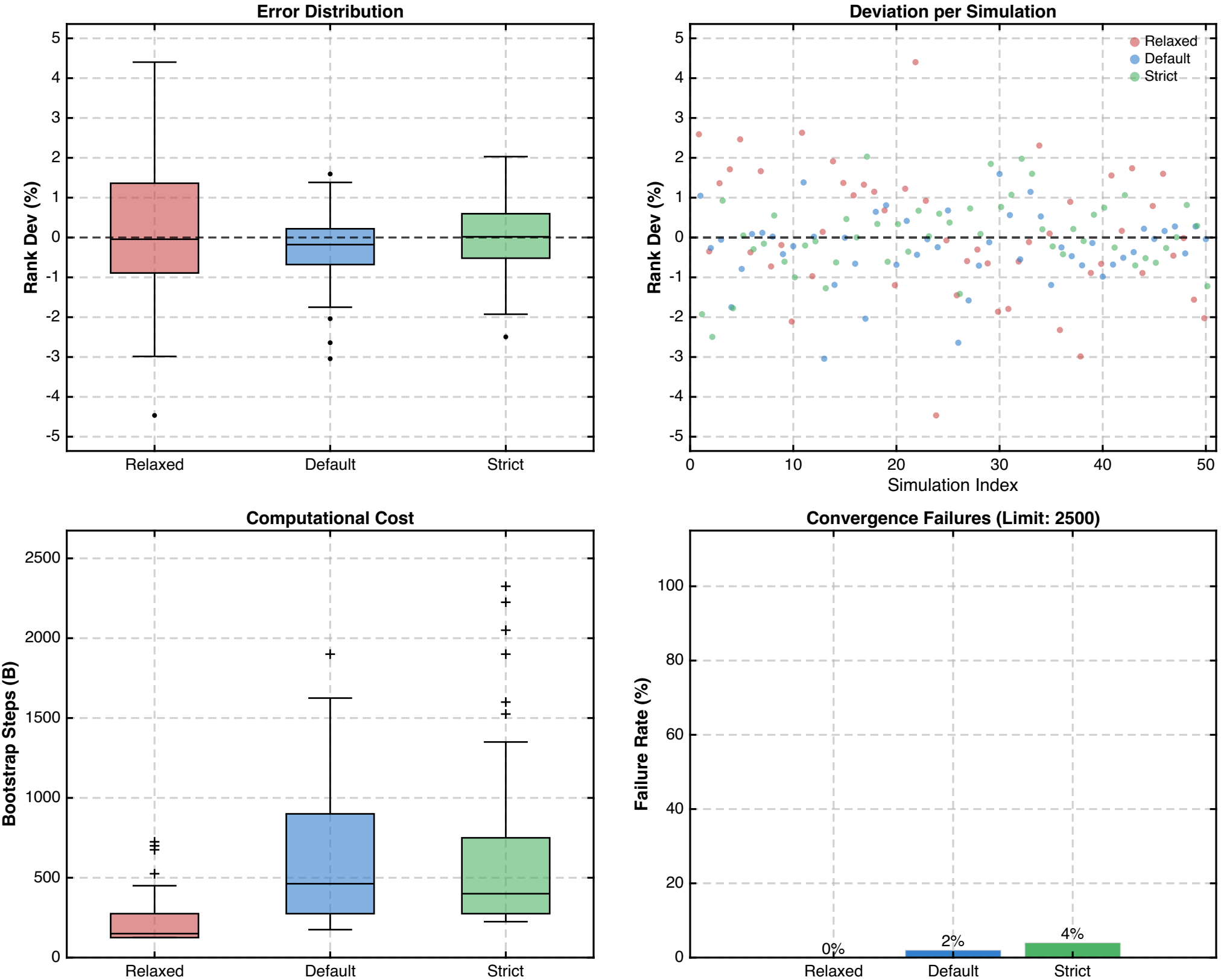
Thresholds (Delta) Analysis: Sensitivity (Median Cliff's d = 0.18)



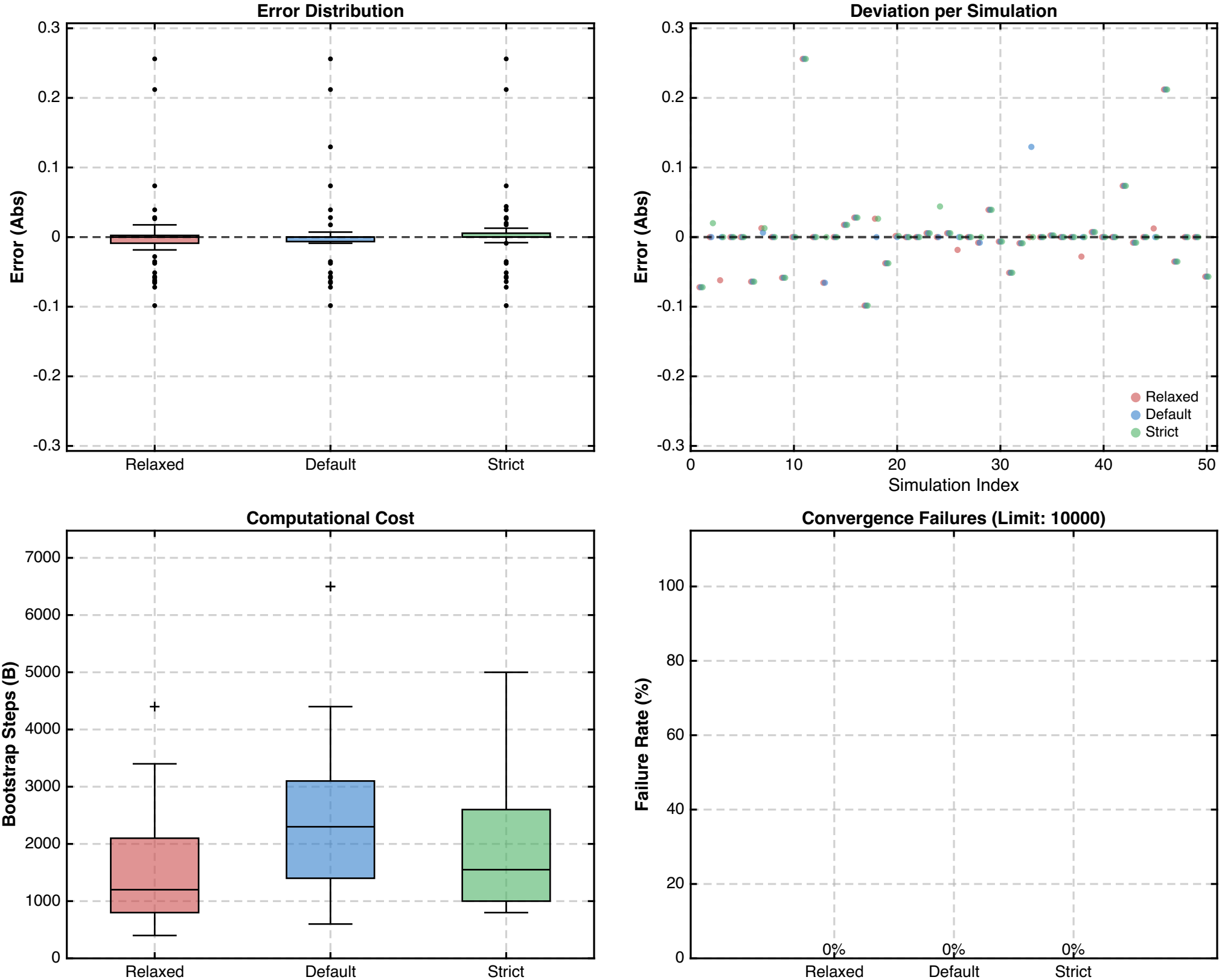
BCa CI (Width) Analysis: Sensitivity (Median Cliff's d = 0.18)



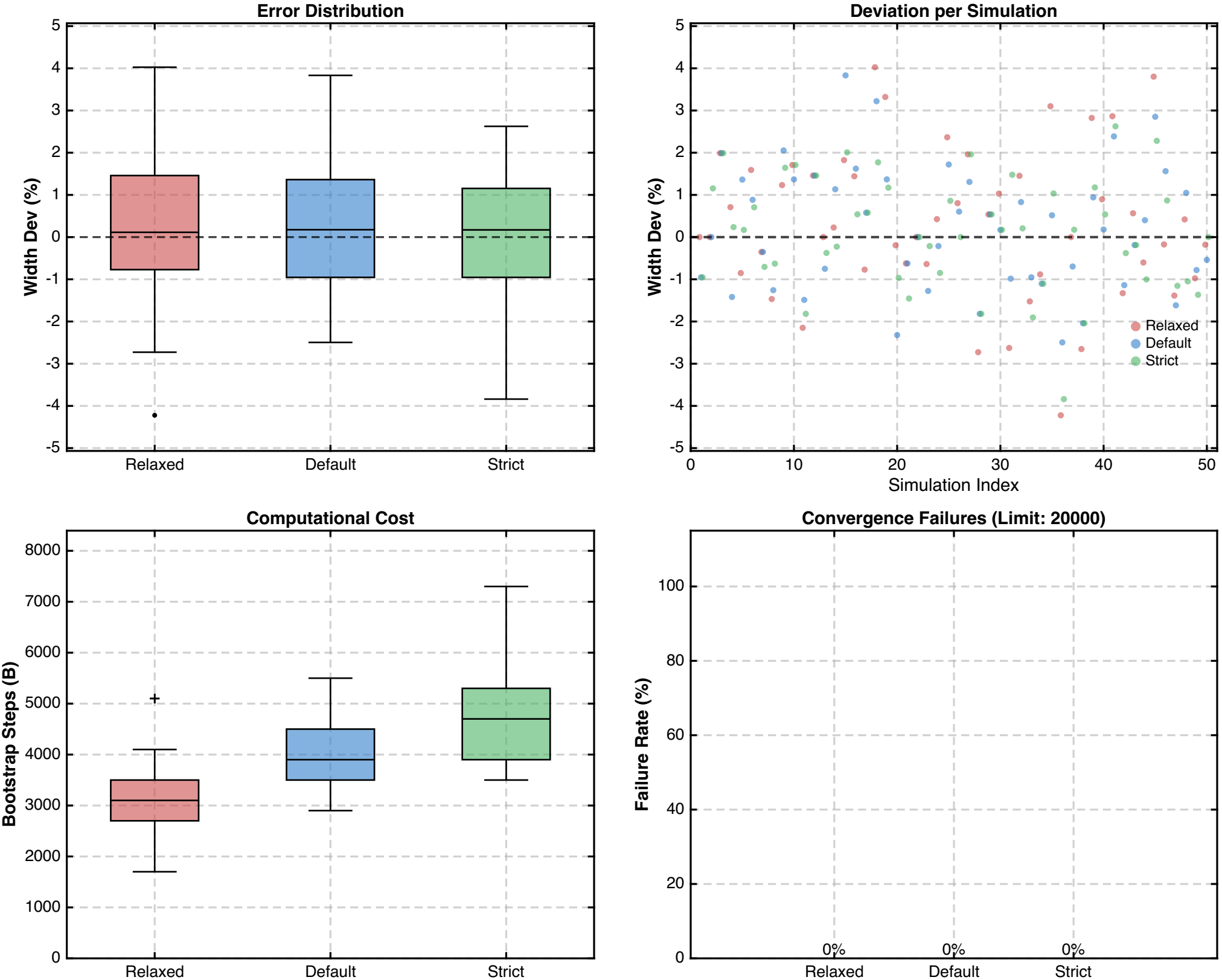
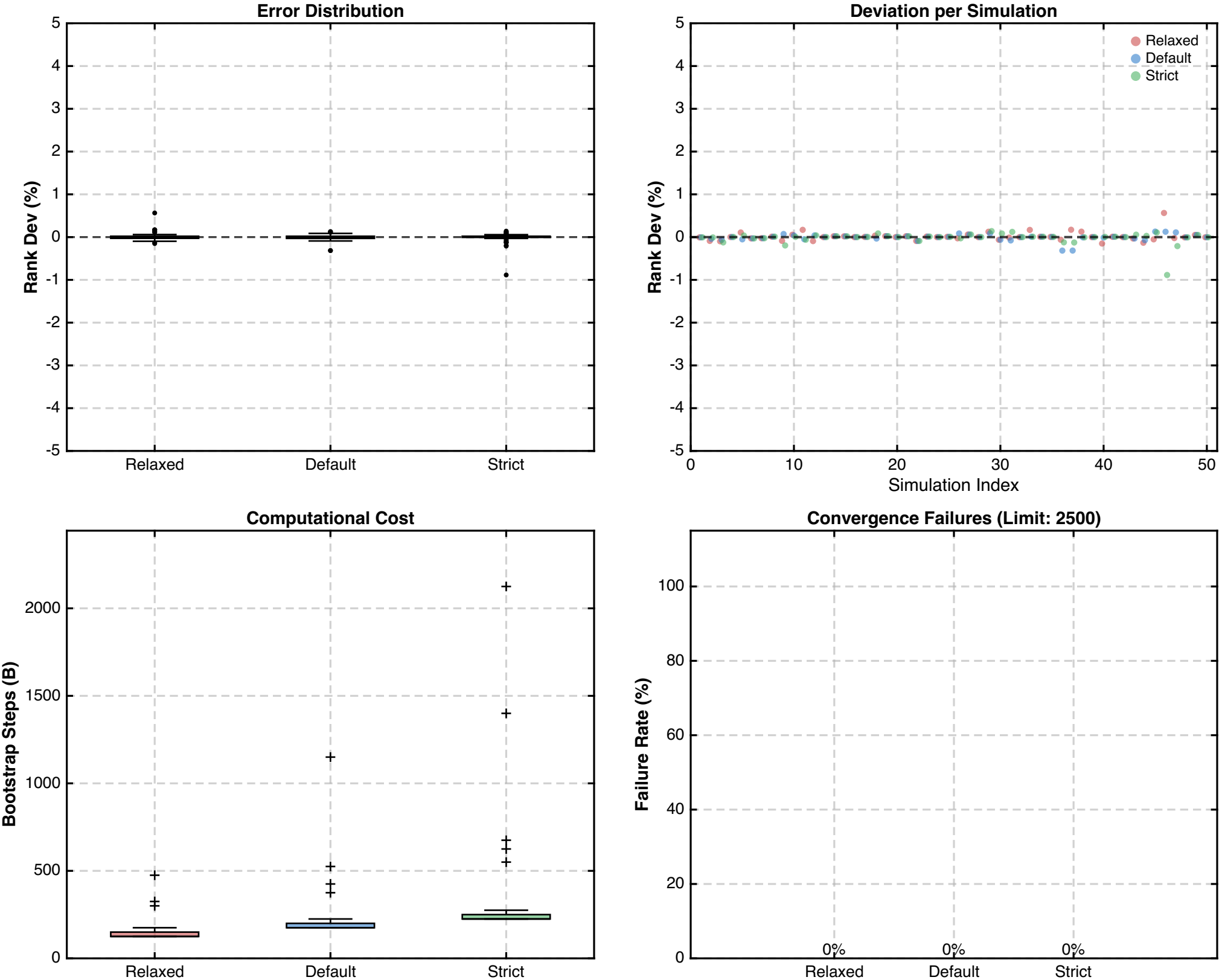
Ranking (Mean) Analysis: Sensitivity (Median Cliff's d = 0.18)



Thresholds (Delta) Analysis: Discrimination (Median Cliff's d = 0.74)



BCa CI (Width) Analysis: Discrimination (Median Cliff's d = 0.74)



Ranking (Mean) Analysis: Discrimination (Median Cliff's d = 0.74)